

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 16

Skin

Chapter Goals (Slide 1 of 2)

- Name the layers of the skin and the accessory structures associated with the skin.
- Build medical words using the combining forms that are related to the specialty of dermatology.
- Identify lesions, signs, and symptoms, and pathologic conditions that relate to the skin.

Chapter Goals (Slide 2 of 2)

- Describe laboratory tests and clinical procedures that pertain to the skin and recognize relevant abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 16

Lesson 16.1

Introduction

Skin: integumentary system

- Weighs 8 to 10 lbs.
- Covers 22 sq. ft. in the average adult

Functions of Skin (Slide 1 of 2)

- Provides **protective** membrane
- Skin glands **lubricate and cool** the skin
- **Receptor** for sensations
- Helps maintain body **temperature**

Functions of Skin (Slide 2 of 2)

Produces sweat: sweat glands produce a watery secretion that evaporates and cools

Produces sebum: sebaceous glands produce an oily secretion that lubricates the skin and hair

Receives sensation: pain, temperature, pressure, and touch

Thermoregulates: interprets message from heat center in the brain

Structure of the Skin (Slide 1 of 3)

Epidermis: outermost, thin cellular membrane

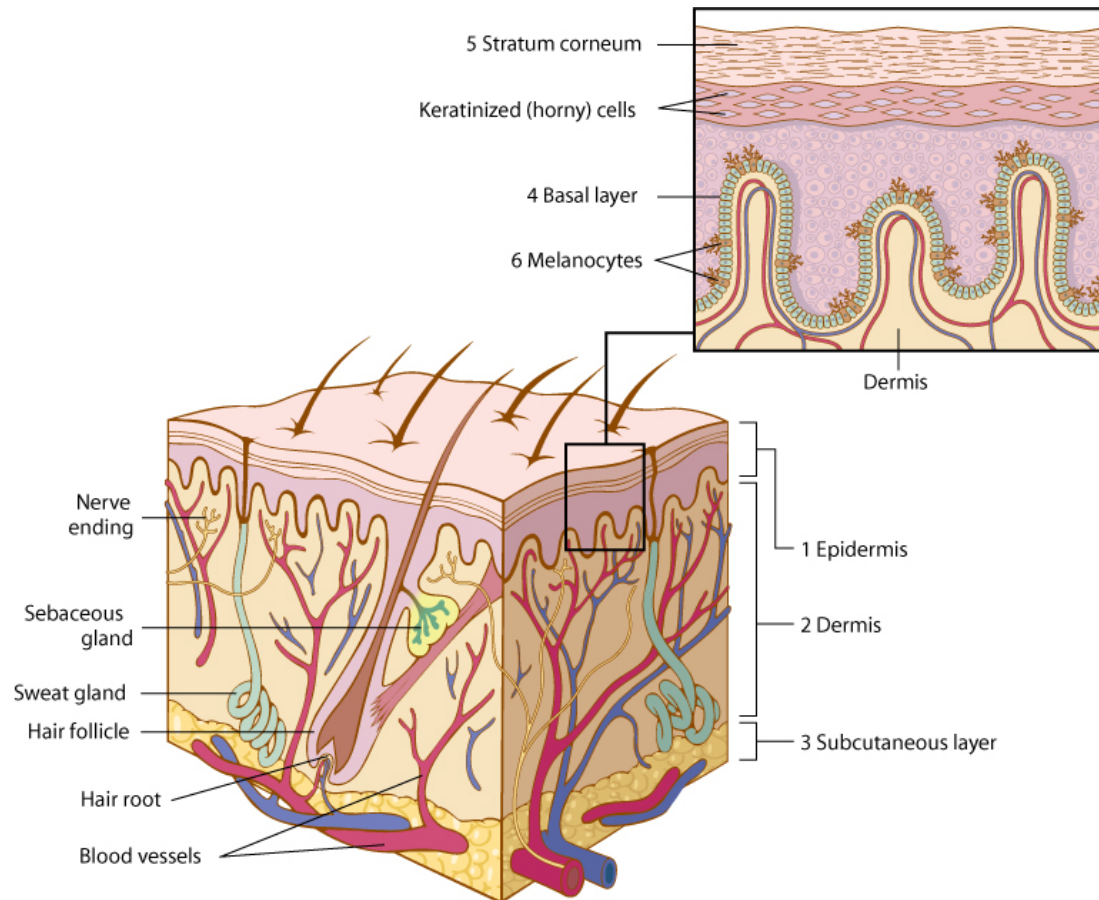
Dermis: next layer; dense fibrous, connective tissue

Subcutaneous tissue: thick, fat-containing tissue

Structure of the Skin (Slide 2 of 3)

- Epidermis layer contains:
- Basal layer
- Stratum corneum
- Melanocytes

Structure of the Skin (Slide 3 of 3)



Accessory Organs of the Skin:

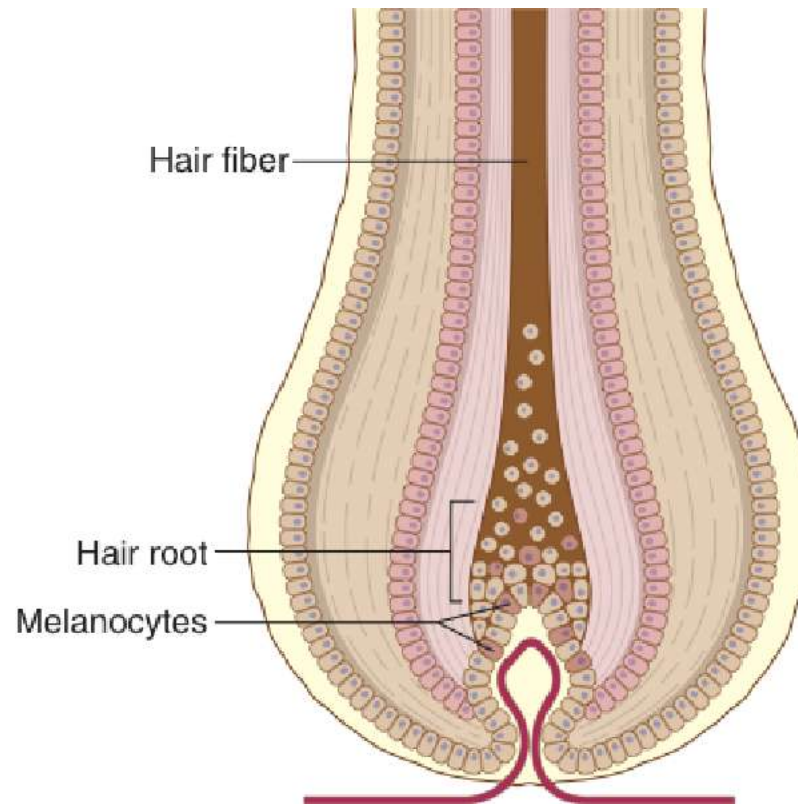
Hair (Slide 1 of 2)

Hair cells are filled with the hard protein; keratin

- **Hair follicles:** shafts that hold the hair
- Five million hairs on the body; 100,000 on the head
- Melanocytes at the root form the color
- Grows .5 inch (1.3 cm) per month
- Cutting does not affect growth

Accessory Organs of the Skin:

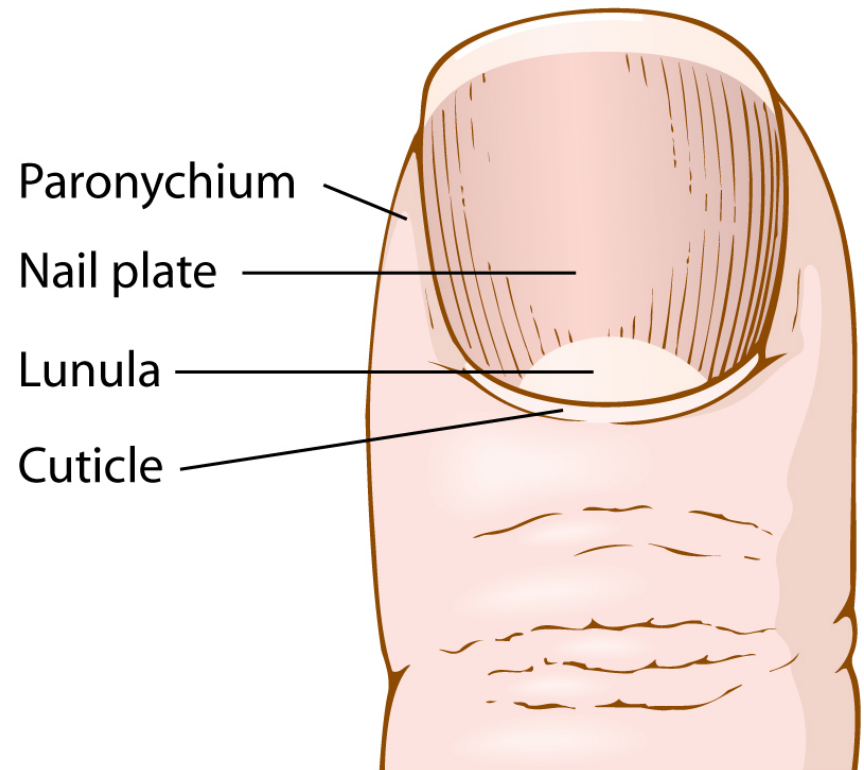
Hair (Slide 2 of 2)



Accessory Organs of Skin: Nails

Hard keratin plates covering the
toes and fingers

- Lunula
- Cuticle
- Paronychium



Accessory Organs of Skin:

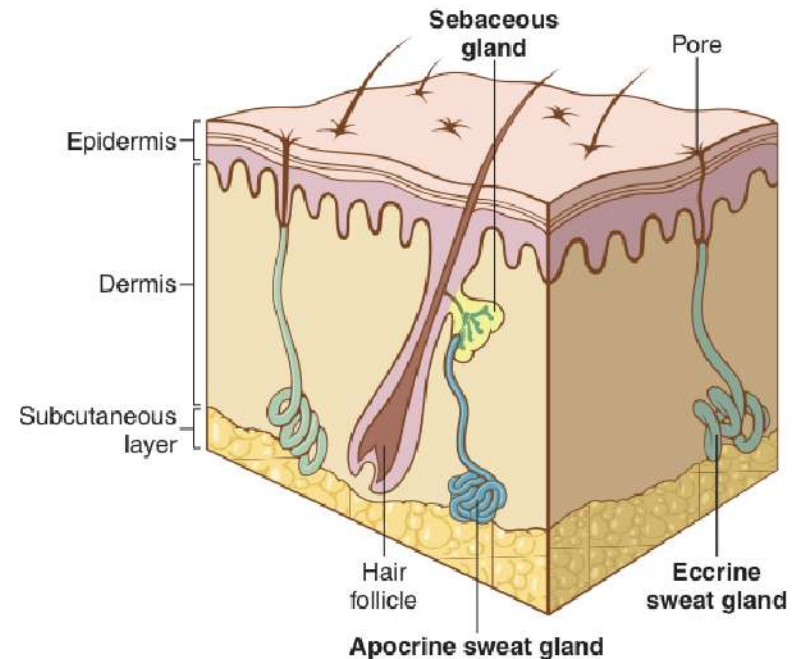
Glands (Slide 1 of 2)

Glands: sebaceous and sweat

- Sebaceous glands secrete oily sebum into the hair follicle to lubricate.
- Sweat glands secrete into pores to moisten and cool.
- Both are subject to bacterial growth.

Accessory Organs of Skin: Glands (Slide 2 of 2)

Sebaceous gland, eccrine sweat gland, and apocrine sweat gland



Vocabulary (Slide 1 of 16)

Term
adipocyte
albino
apocrine sweat gland

Vocabulary (Slide 2 of 16)

Term	Meaning/Definition
adipocyte	Fat cell
albino	Person with skin deficient in pigment
apocrine sweat gland	One of the large dermal exocrine glands located in the axilla and genital areas

Vocabulary (Slide 3 of 16)

Term
basal layer
collagen
cuticle

Vocabulary (Slide 4 of 16)

Term	Meaning/Definition
basal layer	Deepest region of the epidermis; gives rise to all the epidermal cells
collagen	Structural protein found in the skin and connective tissue
cuticle	Band of epidermis at the base and sides of the nail plate

Vocabulary (Slide 5 of 16)

Term
dermis
ecocrine sweat gland
epidermis

Vocabulary (Slide 6 of 16)

Term	Meaning/Definition
dermis	Middle layer of the skin
eccrine sweat gland	Most numerous sweat-producing exocrine gland in skin
epidermis	Outermost layer of the skin

Vocabulary (Slide 7 of 16)

Term
epithelium
hair follicle
integumentary system

Vocabulary (Slide 8 of 16)

Term	Meaning/Definition
epithelium	Layer of skin cells forming the outer and inner surfaces of the body
hair follicle	Sac within which each hair grows
integumentary system	The skin and its accessory structures such as hair and nails

Vocabulary (Slide 9 of 16)

Term
keratin
lunula
melanin

Vocabulary (Slide 10 of 16)

Term	Meaning/Definition
keratin	Hard protein material found in the epidermis, hair, and nails
lunula	Half-moon-shaped, whitish area at the base of a nail
melanin	Skin pigment formed by melanocytes in the epidermis

Vocabulary (Slide 11 of 16)

Term
paronychium
pore
sebaceous gland

Vocabulary (Slide 12 of 16)

Term	Meaning/Definition
paronychium	Soft tissue surrounding the nail border
pore	Tiny opening on the surface of the skin
sebaceous gland	Oil-secreting gland in the dermis that is associated with hair follicles

Vocabulary (Slide 13 of 16)

Term
sebum
squamous epithelium
stratified

Vocabulary (Slide 14 of 16)

Term	Meaning/Definition
sebum	Oily substance secreted by sebaceous glands
squamous epithelium	Flat, scale-like cells composing the epidermis
stratified	Arranged in layers

Vocabulary (Slide 15 of 16)

Term
stratum (<i>plural: strata</i>)
stratum corneum
subcutaneous layer

Vocabulary (Slide 16 of 16)

Term	Meaning/Definition
stratum (<i>plural:</i> strata)	A layer (of cells)
stratum corneum	Outermost layer of the epidermis; consists of flattened, keratinized cells
subcutaneous layer	Innermost layer of the skin, containing fat tissue

Combining Forms (Slide 1 of 5)

Combining Form	Meaning
adip/o	fat
albin/o	white
caus/o	burn, burning
cauter/o	heat, burn
cutane/o	skin
derm/o	skin

Combining Forms (Slide 2 of 5)

Combining Form	Meaning
dermat/o	skin
diaphor/o	profuse sweating
erythem/o	redness
erythemat/o	redness
hidr/o	sweat
ichthy/o	scaly, dry

Combining Forms (Slide 3 of 5)

Combining Form	Meaning
kerat/o	hard, horny tissue
leuk/o	white
lip/o	fat
melan/o	black
myc/o	fungus
onych/o	nail

Combining Forms (Slide 4 of 5)

Combining Form	Meaning
phyt/o	plant
pil/o	hair, hair follicle
py/o	pus
rhythid/o	wrinkle
seb/o	sebum
squam/o	scale-like

Combining Forms (Slide 5 of 5)

Combining Form	Meaning
steat/o	fat
trich/o	hair
ungu/o	nail
xanth/o	yellow
xer/o	dry

QUICK QUIZ (Slide 1 of 2)

1. Which combining form refers to white?

A. chlor/o

B. jaund/o

C. melan/o

D. albin/o

QUICK QUIZ (Slide 2 of 2)

- **2. Which combining form refers to the same color as jaund/o?**
 - A. xanth/o
 - B. chlor/o
 - C. erythr/o
 - D. cyan/o

Colors

TABLE 16-1 Colors

Combining Form	Meaning	Terminology
albin/o	white	<u>albin</u> ism
anthrac/o	black (as coal)	<u>anthrac</u> osis
chlor/o	green	<u>chlor</u> ophyll
cirrh/o	tawny yellow	<u>cirrh</u> osis
cyan/o	blue	<u>cyan</u> osis
eosin/o	rosy	<u>eosin</u> ophil
erythr/o	red	<u>erythr</u> ocyte
jaund/o	yellow	<u>jaund</u> ice
leuk/o	white	<u>leuk</u> oderma
lute/o	yellow	corpus <u>luteu</u> m
melan/o	black	<u>melan</u> ocyte
poli/o	gray	<u>poli</u> omyelitis
xanth/o	yellow	<u>xanth</u> oma

Chapter 16

Lesson 16.2

Cutaneous Lesions

- Types of cutaneous lesions:
 - Crust (scab)
 - Cyst
 - Erosion
 - Fissure
 - Macule
 - Nodule
 - Papule
 - Pustule
 - Ulcer
 - Vesicle
 - Wheal

Signs and Symptoms (Slide 1 or 4)

Alopecia: the absence of hair where it normally grows



Signs and Symptoms (Slide 2 of 4)

Ecchymosis: blue-purple marks on the skin



Signs and Symptoms (Slide 3 of 4)

Petechia: small pinpoint hemorrhage



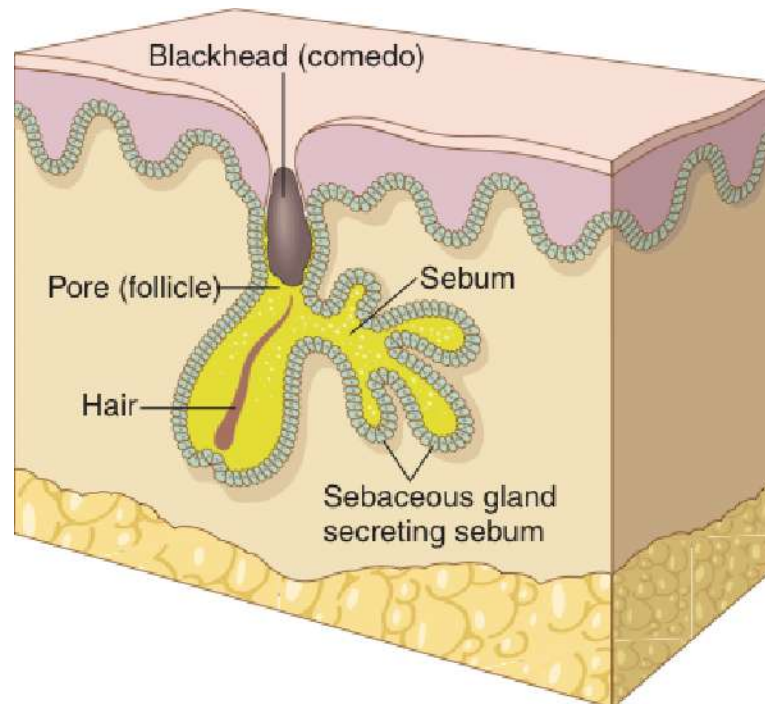
Signs and Symptoms (Slide 4 of 4)

Urticaria: an acute allergic reaction with red, round wheals on the skin



Abnormal Conditions (Slide 1 of 7)

Acne: papular and pustular eruption of skin with increased production of sebum



Abnormal Conditions (Slide 2 of 7)

Burns: injury to tissue due to heat, chemical, electric shock, lightning, or radiation. Second degree burn shown in figure.



Abnormal Conditions (Slide 3 of 7)

- **Cellulitis:** a diffuse acute infection of the skin
- **Eczema (atopic dermatitis):** inflammation of the skin with erythematous and papulovesicular lesions caused by allergy
- **Exanthematous viral diseases:** a rash due to virus (for example, rubella)
- **Gangrene:** the death of tissue with loss of blood supply

Abnormal Conditions (Slide 4 of 7)

- **Impetigo:** contagious pyoderma caused by staph or strep
- **Psoriasis:** chronic recurrent dermatosis with silver gray scales that itch
- **Scabies:** parasitic (tiny mites) and infectious pruritus
- **Scleroderma:** chronic and progressive disease of the skin with hardening of connective tissue

Abnormal Conditions (Slide 5 of 7)

- **Systemic lupus erythematosus (SLE):** an inflammatory disease of collagen in the skin, joints, and internal organs
- **Urticaria (hives):** an acute allergic reaction in which red, round wheals develop on the skin

Abnormal Conditions (Slide 6 of 7)

- **Tinea:** infection of the skin caused by a fungus— Tinea corporis shown in figure



Abnormal Conditions (Slide 7 of 7)

Vitiligo: loss of pigment in areas of the skin



Chapter 16

Lesson 16.3

Skin Neoplasms — Benign

- **Keratosis:** thickened area of epidermis
- **Leukoplakia:** white, thickened patches on tongue or cheek
- **Nevus (*pl. nevi*):** pigmented lesion
- **Verruca:** warts caused by virus

Skin Neoplasms — Cancerous

(Slide 1 of 3)

Basal cell carcinoma
— Malignant tumor
of the basal cell
layer of the
epidermis

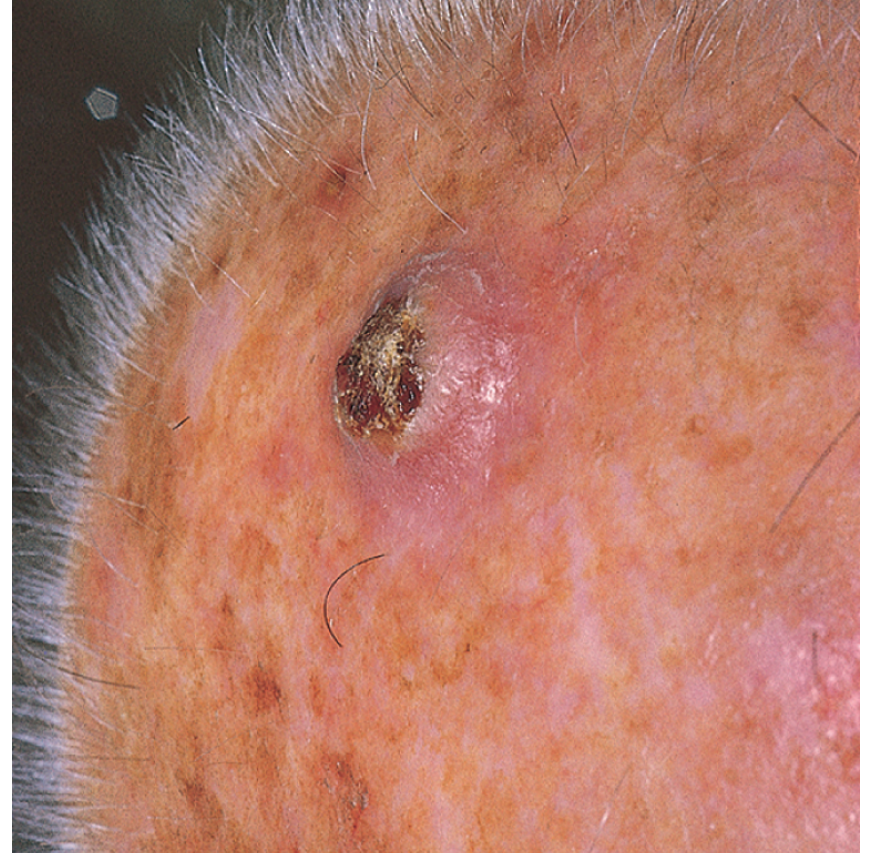


Skin Neoplasms — Cancerous

(Slide 2 of 3)

Squamous cell carcinoma--

Malignant tumor of the squamous epithelial cells of the epidermis



Skin Neoplasms — Cancerous

(Slide 3 of 3)

Kaposi sarcoma

Malignant, vascular, neoplastic growth
characterized by cutaneous nodules

Laboratory Tests

Bacterial analyses: pus or fluid samples examined to detect microorganisms

Fungal tests: scrapings for culture and microscopic examination after treatment with KOH

Clinical Procedures (Slide 1 of 2)

Cryosurgery: destroy tissue with subzero temperatures using liquid nitrogen

Curettage: scrape lesion with sharp curet

Electrodesiccation: destroy tissue by burning with electric spark

Clinical Procedures (Slide 2 of 2)

Mohs surgery: removes thin layers of growth to examine under a microscope (basal and squamous cell)

Skin biopsy: punch and shave to remove for examination in path lab

Skin test: test reaction of body to allergen with skin test (scratch or patch tests)

Abbreviations (Slide 1 of 2)

Abbreviation

ABCDE

Bx

Derm.

DLE

PPD

PUVA

SLE

SC

Abbreviations (Slide 2 of 2)

Abbreviation	Meaning
ABCDE	asymmetry, border, color, diameter, evolution (or change) —characteristics associated with melanoma
Bx	biopsy
Derm.	dermatology
DLE	discoid lupus erythematosus
PPD	purified protein derivative
PUVA	psoralen-ultraviolet A light therapy
SLE	systemic lupus erythematosus
SC	subcutaneous

Review Sheet – Combining Forms

(Slide 1 of 10)

Combining Form
adip/o
albin/o
caus/o
cauter/o
cutane/o
derm/o

Review Sheet – Combining Forms

(Slide 2 of 10)

Combining Form	Meaning
adip/o	fat
albin/o	white
caus/o	burn, burning
cauter/o	heat, burn
cutane/o	skin
derm/o	skin

Review Sheet – Combining Forms

(Slide 3 of 10)

Combining Form
dermat/o
diaphor/o
erythem/o
erythemat/o
hidr/o
ichthy/o

Review Sheet – Combining Forms

(Slide 4 of 10)

Combining Form	Meaning
dermat/o	skin
diaphor/o	profuse sweating
erythem/o	redness
erythemat/o	redness
hidr/o	sweat
ichthy/o	scaly, dry

Review Sheet – Combining Forms

(Slide 5 of 10)

Combining Form
kerat/o
leuk/o
lip/o
melan/o
myc/o
onych/o

Review Sheet – Combining Forms

(Slide 6 of 10)

Combining Form	Meaning
kerat/o	hard, horny tissue
leuk/o	white
lip/o	fat
melan/o	black
myc/o	fungus
onych/o	nail

Review Sheet – Combining Forms

(Slide 7 of 10)

Combining Form
phyt/o
pil/o
py/o
rhythid/o
seb/o
squam/o

Review Sheet – Combining Forms

(Slide 8 of 10)

Combining Form	Meaning
phyt/o	plant
pil/o	hair, hair follicle
py/o	pus
rhytid/o	wrinkle
seb/o	sebum
squam/o	scale-like

Review Sheet – Combining Forms

(Slide 9 of 10)

Combining Form
steat/o
trich/o
ungu/o
xanth/o
xer/o

Review Sheet – Combining Forms

(Slide 10 of 10)

Combining Form	Meaning
steat/o	fat
trich/o	hair
ungu/o	nail
xanth/o	yellow
xer/o	dry

Review Sheet – Suffixes (Slide 1 of 4)

Suffix
-algia
-derma
-esis
-lysis
-ose

Review Sheet – Suffixes (Slide 2 of 4)

Suffix	Meaning
-algia	pain
-derma	skin
-esis	condition
-lysis	breakdown; separation; destruction; loosening
-ose	full of; pertaining to; sugar

Review Sheet – Suffixes (Slide 3 of 4)

Suffix
-osis
-ous
-plakia
-plasty
-rrhea

Review Sheet – Suffixes (Slide 4 of 4)

Suffix	Meaning
-osis	condition, usually abnormal
-ous	pertaining to
-plakia	plaque
-plasty	surgical repair
-rrhea	flow; discharge